Examiner only

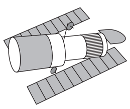
3. *(a)* Explain why the age of the Universe can be approximated as $\frac{1}{H_0}$, where H_0 is the Hubble constant. [2]

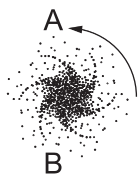
(b) The megaparsec (Mpc) is a unit of astronomical distance equal to 3.09×10^{22} m. Use Hubble's law to show that the expected redshift for a supernova at a distance of 1 Mpc for a wavelength of 486.1 nm is approximately 0.11 nm. [3]

(c) The spiral galaxy shown is rotating anticlockwise and is viewed by the Hubble Space Telescope.

Images not to scale

Hubble Space Telescope





Rotating spiral galaxy

The measured **blue** shift at point A of the galaxy is 0.22 nm (for the 486.1 nm wavelength) and the measured **redshift** at point B of the galaxy is 0.66 nm (for the same wavelength). Calculate the recessional velocity of the galaxy and the rotational speed of the galaxy at A (assume that A and B have the same rotational speeds). [5]

Examiner only

(d) Explain how spiral galaxies provide evidence for the existence of dark matter and how this evidence has been gathered. [6 QER]

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